

Research paper

Techno-economic study of different process configurations of CO₂ capture by MEA absorption and utilization of an MCDM method for their ranking

Mohsen Salimi^a, Vahid Karimi^b, Farzin Hosseinfard^b, Majid Amidpour^{b,*}

^a Renewable Energy Research Department, Niroo Research Institute (NRI), Tehran, Iran

^b Faculty of Mechanical Engineering, Department of Energy System Engineering, K.N. Toosi University of Technology, Tehran, Iran

ARTICLE INFO

Keywords:

Absorption
Carbon capture
Configuration
Monoethanolamine
Post-combustion
Techno-economic

ABSTRACT

Carbon dioxide emissions from fossil fuel-fired power stations are attracting growing attention as a potential approach for reducing greenhouse gas emissions. Post-combustion capture of CO₂ with monoethanolamine (MEA) is one of the most successful and mature technologies for mitigating carbon dioxide emissions. The most crucial problem of this technique is the high heat requirement of the reboiler for solvent regeneration. In this study, seven different configurations for carbon capture, namely, standard absorption, vapor recompression, vapor recompression process with split-flow, absorption column with intercooling, rich-split, combined, and rich-split with vapor recompression, are studied. The simulation results showed that the reboiler energy consumption intensity of alternative configurations lowered by 15.2 %, 18.4 %, 2.9 %, 7.4 %, 20.3 %, and 23.2 %, respectively, with regard to the standard absorption configuration. Seven configurations were ranked based on the total investment cost, steam consumption, power consumption, and water consumption. Combined and rich-split were the best, and vapor recompression and Rich-split with vapor recompression were the worst.

1. Introduction

1.1. Background

Anthropogenic emissions have dramatically increased greenhouse gas concentration in the atmosphere. The primary source of carbon dioxide as the most important GHG is the combustion of fossil fuels in power plants. Fossil fuels provide around 81 % of the global primary energy supply (KEY WORLD ENERGY STATISTICS, 2019). Decreasing global energy systems' dependency on fossil fuel energy sources is challenging (Ziemele et al., 2023). Increased carbon emissions following the Industrial Revolution have led to numerous issues, including rising sea levels (Qu et al., 2020), global warming, adverse health effects on humans, and acid rain (Grennfelt et al., 2020). To address these challenges, various agreements have been established by governments, such as the 1992 Rio Agreement in Brazil (Milhorance et al., 2022), the 1997 Kyoto Protocol in Japan (Depledge, 2022), the 2001 Morocco Agreement, and most notably, the 2015 Paris Agreement (Sun et al., 2022). The Paris Agreement aims to limit global warming to 1.5 degrees Celsius by the end of the century. In recent years, the use of carbon capture technology has seen a significant increase as a solution to combat this

issue. Carbon capture technology comes in various forms, including adsorption (Liu et al., 2024), absorption (Hosseinfard et al., 2024a), and membrane absorption (Soeppan et al., 2024). Commercially available amines like MEA (Zhou et al., 2023), DGA (Hosseinfard et al., 2024b), and others are used to remove carbon dioxide from flue gas. Among the different methods of carbon capture technology, post-combustion is the most commonly used due to its compatibility with existing power plants. The energy efficiency of power plants should be increased, energy demand should be reduced, and CO₂ capture and storage (CCS) should be commercialized to reduce carbon dioxide emissions of fossil fuel utilization. Retrofitting existing power plants with CCS is one of the primary methods to mitigate CO₂ emissions in energy-consuming sectors. Absorption processes based on MEA are commercial technologies for capturing CO₂ from the flue gas of power plants. Post-combustion carbon capture can be optimized using methods like Multiple Criteria Decision Making (MCDM). MCDM is a systematic approach used in decision-making processes involving multiple criteria, helping to select the best option among several configurations (Alsalem et al., 2022a). This method considers various criteria and their weights, aiding decision-makers in improving their performance. MCDM is commonly applied in diverse engineering fields (Ismael et al., 2023), including energy (Sotiropoulou and Vavatsikos, 2021), environmental

* Corresponding author.

E-mail address: amidpour@kntu.ac.ir (M. Amidpour).

<https://doi.org/10.1016/j.egy.2024.06.043>

Received 21 April 2024; Received in revised form 30 May 2024; Accepted 17 June 2024

Available online 27 June 2024

2352-4847/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

Nomenclature		MDEA	Methyl diethanolamine
Symbols	A	MEA	Monoethanolamine
	a_f	NO_x	Nitric oxide
	C	NPV	Net Present Value
	CCS	PCC	Post Combustion Carbon Capture
	C_f	PROMETHEE	Preference Ranking Organization Method for Enrichment Evaluation
	DC	PW	Present Worth
	DGA	PZ	Piperazine
	ELECTRE	Q_h	Thermal Duty
	FCI	SO_x	Sulfur oxide
	GHG	T_c	Sink Temperature
	h_{bw}	TCI	Total Cost of Investment
	H_{LP}	T_h	Source Temperature
	HX	TOPSIS	Technique for Order Performance Similarity to Ideal Solution
	HYSIM	USD	United state dollars
	IDC	V	Purchase (\$)
	LCC	W_c	Work of compressor
	LVC	W_{eq}	Work of Equal
	m	WI	Working Investment
	MAUT	W_p	Work of Pumps
	MCDM	wt	weight
	η		

management (Akbari et al., 2021), economics (Mohammed et al., 2019), and health (Alsalem et al., 2022b).

1.2. Literature review

Jung et al (Jung et al., 2013). studied the CO₂ capture process configuration based on MEA scrubbing. A configuration of the MEA process based on split flow and phase-separation heat exchanger was introduced to reduce the reboiler energy requirement. The phase-separation heat exchanger decreases preheating duty loss, and the split flow reduces the reflux ratio of the stripper. Therefore, the required regeneration energy of the presented configuration is 27 % lower than the reference configuration. Choi et al (Choi et al., 2019). worked on the process design of MEA-based carbon dioxide capture processes for GHG emission-intensive industries. In different sectors, flue gases have other conditions. The concentrations of CO₂ in the flue gases of various sectors are significant and have a crucial impact on the design of CO₂ capture processes. This paper presents a methodology for optimizing a superstructure for different flue gas compositions found in industries such as steel, cement, and refinery plants. Optimization of operating conditions and structural modifications leads to the optimal configurations for different industrial emitters of carbon dioxide. Liu (Liu, 2020) used cold-split bypass, interheated stripper, and lean vapor compression in stripper configurations as energy-saving mechanisms. In two studies (Lin et al., 2014; Le Moulec et al., 2014), the cold-split bypass modification was confirmed for retrieving vapor heat from the flow directly entering the condenser. The interheater was utilized for adjusting the temperature profile of the desorber by increasing the temperature of its bottom section, which helps carbon dioxide desorption (Karimi et al., 2012). This study proposed a combination of the cold-split bypass and interheater to enhance their advantages and energy saving. With a higher carbon dioxide purity at the top of the desorber, a lower rate of overhead vapor is required to be produced. On the other hand, the pressurized desorber's elevated temperature accelerates the MEA solvent's thermal degradation as the reboiler's temperature goes beyond 135 °C. Oh et al (Oh et al., 2016). studied the energy minimization of the carbon dioxide capture process based on the MEA. This study presented a novel superstructure for optimizing carbon dioxide capture processes.

The introduced methodology was applied to a case study to show the effectiveness of the proposed framework in evaluating available design options for upgrading energy efficiency. Dubois and Thomas (Dubois and Thomas, 2018) compared different configurations of the carbon dioxide capture process utilizing various solvents for flue gases of cement plants. Considerable energy savings were achieved by utilizing other solvents. Rich vapor compression configuration utilizing piperazine (PZ)-methyl diethanolamine (MDEA) solvent has the best energy performance. Ahn et al (Ahn et al., 2013). modeled various configurations of MEA-based plants of CO₂ capture integrated coal-fired power plants. The configurations were compared based on the total electrical and thermal energy consumptions to capture 90 % of CO₂ from the sub-critical pulverized coal boiler power plant. Several modifications to the base configuration reduced the boiler duty more than a single modification. In this study, we utilize the Multiple Criteria Decision Making (MCDM) method to address issues and enhance the quality of planning. MCDM has been increasingly employed in recent years and was also utilized by researchers in the 1900s, including methods such as ELECTRE (Elimination and Choice Translating Reality), PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluation), and MAUT (Multi-Attribute Utility Theory) (Massam, 1988). Presently, this method has evolved through algorithms like TOPSIS, Simple Additive Weighting (Vafaei et al., 2021), Analytical Network Process, and others. Among these algorithms, TOPSIS is particularly prevalent due to its straightforward process, ease of implementation, and ranking based on the impact of response factors. Compared to other algorithms, TOPSIS demonstrates high accuracy in achieving positive responses within the same number of calculation steps.

1.3. Objective

The energy requirement of the capture process is high and, therefore, costly. Consequently, it is crucial to study the different configurations of the MEA process to decrease the heat requirement. Also, other parameters, such as water consumption, power consumption, and investment costs, should be considered. In this study, seven different configurations of the MEA absorption process were studied technically and economically. In previous studies in the literature, all of the configurations were

not studied along with each other in one study. Then by considering various criteria (total cost of investment, steam consumption, power consumption, water consumption), different configurations were ranked using the TOPSIS method, which is another novelty of the current study. We are utilizing different configurations based on MEA for Post-Combustion Carbon Capture (PCC) to identify the most economically viable option. This analysis is being conducted using Aspen HYSYS V10. These configurations are being compared based on utility consumption. For instance, in the lean vapor recompression approach, we include a compressor to reduce energy consumption in the reboiler. However, this particular configuration increases the investment cost in the plant and the required utility. We are evaluating various configurations to ultimately determine the most cost-effective solution. Compared to other studies in this study, the TOPSIS algorithm within the MCDM methodology was used to select optimal configurations for post-combustion carbon capture. This method considers a wide range of parameters and criteria, providing higher accuracy than traditional methods. Compared to exergoeconomic analysis, TOPSIS can evaluate multiple criteria simultaneously, offering a holistic assessment of alternatives. TOPSIS ranks alternatives systematically, aiding in well-rounded decision-making. It is simpler to implement than exergoeconomic analysis and accessible to a broader user base also the combination of various configurations in the study showcased benefits, such as reduced solvent regeneration. Overall, TOPSIS with MCDM presents a comprehensive and efficient approach to decision-making in applications like post-combustion carbon capture.

2. Methodology

These configurations should be simulated to compare various carbon capture configurations mentioned in the previous section from the technical point of view. Therefore, these configurations are simulated with Aspen HYSYS v10. The most critical parameter for carbon capture process selection is the carbon dioxide concentration in the exhaust flue gas of emitter units. The carbon dioxide concentration of most of Iran’s power plants is 3–4 molar percent of the flue gas. In this study, the following assumptions were made

- The carbon capture rate is 85 %.
- The amine concentration is 30 %.
- The model operates under steady-state conditions.
- The energy and mass balance equations are formulated in one dimension.
- The flue gas specifications can be found in Table 1.
- Components in the flue gas in parts per million (ppm) were neglected in the analysis.
- Pumps, Compressors, and turbine efficiency is 75 %.
- The type of absorber and stripper column is a packed column.

The power plant’s flue gas composition, flow rate, temperature, and pressure entering the capture unit are presented in Table 1. This study assumes that the flue gas’s SO_x and NO_x were removed before the carbon capture unit.

The thermodynamic equation used in Aspen HYSYS v10 for the simulation of this study is the Acid Gas Property Package. HYSIM Inside-

Out solver was used for its proper convergence of absorption and desorption columns. Fig. 1 illustrates the methodology path followed in this study. Initially, various configurations were simulated using Aspen HYSYS to reduce energy consumption. Subsequently, the TOPSIS algorithm was employed to determine the optimal configuration among the simulated ones by evaluating the group matrix performance based on economic parameters. The rankings and comparisons of the configurations were then characterized and exported using the softmax function to convert vector to value. TOPSIS is a multi-criteria decision-making method employed in this study for comparing various configurations in post-combustion carbon capture. The fundamental concept of TOPSIS involves determining the alternative that is closest to the ideal solution while being farthest from the negative ideal solution. Initially, criteria are defined to evaluate aspects such as cost and economic gain, followed by listing the different carbon capture configurations in the alternative section. The subsequent calculation involves constructing a decision matrix where rows represent alternatives and columns represent the previously defined criteria. Normalization is then applied to transform values between 0 and 1, followed by the determination of weights for each criterion based on the normalized decision matrix. The final step entails calculating the distance from the ideal situation using the Euclidean distance metric for each alternative.

2.1. Process description

In this section, seven different configurations are studied. These configurations are named standard absorption process, vapor recompression process, lean split with vapor recompression process, absorption column with intercooling process, rich-split process, combined process, and rich-split with vapor recompression process. Their stages and process equipment are explained in brief. The selected solvent in this study is MEA. Its molecular formula is C₂H₇NO. There have been many studies exploring solvents other than MEA. However, MEA is still the most common solvent for carbon dioxide absorption under atmospheric conditions (Lars Erik, 2012). Oxemann and Kather (Oexmann and Kather, 2010) claimed that a solvent like MEA with a high absorption heat might utilize less heat than lower heat absorption solvents at atmospheric pressure, which is a suitable condition for carbon dioxide removal. MEA is one of the best solvents for removing carbon dioxide and hydrogen sulfide (Kohl Richard Nielsen, 1997). High reaction ability, high capture capacity, thermal stability, and acceptable

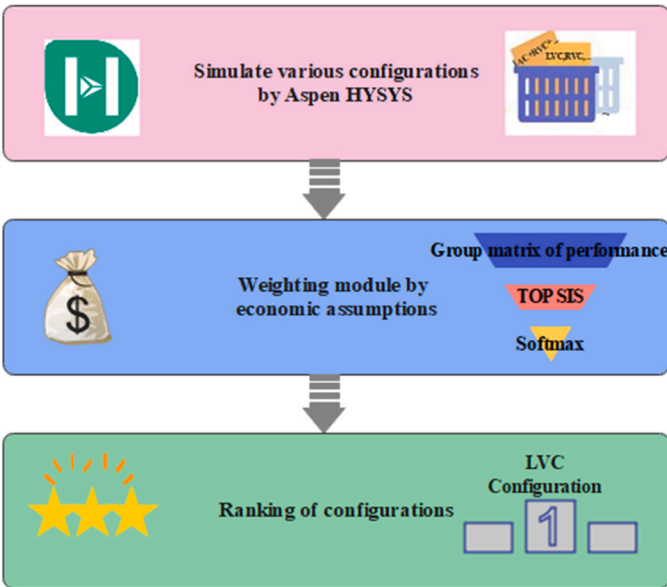


Fig. 1. Configurations selection path.

Table 1

Flue gas composition.

Parameter		
Composition	Water	6.89 mol%
	Nitrogen	75.89 mol%
	Carbon Dioxide	3.72 mol%
	Oxygen	13.5 mol%
Pressure	101 kPa	
Temperature	40 °C	
Mass flow rate	400 kg/s	

desorption rate. High vapor pressure and high reaction heat are its disadvantages, which lead to increased energy demand in the amine desorption stage.

2.2. Standard configuration

The standard absorption configuration is demonstrated in Fig. 2. Performances of alternative configurations are compared to the standard absorption process. This configuration consists of an absorber, a desorber with a reboiler and a condenser, pumps, a cooler, and a heat exchanger. Carbon dioxide from the flue gas of the power plant is absorbed in the absorption column by utilization of monoethanolamine (MEA) as the amine solvent. The carbon-dioxide-rich MEA solution would then be pumped through a rich/lean HX, which will be heated earlier than the stripper (desorber) for regeneration. The reproduced lean amine is pumped again to the absorption column. Then it flows through a rich/lean HX, where its heat is used to raise the temperature of the carbon dioxide-rich flow and cooled in the amine cooler. The standard configuration has the advantage of not requiring additional units, but there is a significant increase in fuel consumption

2.3. Vapor recompression process configuration

In the vapor recompression process, a valve creates a pressure drop for the flashing of regenerated amine from the desorption column bottom. The flashing is the main difference between this configuration and the reference configuration. The produced vapor from the flashing is split from the lean-amine flow using a separator. Then, the vapor is compressed and sent to the desorption column for regeneration. This increases the stripping vapor in the desorber without affecting the water balance of the carbon capture system (Cousins et al., 2011a, 2011b). Fig. 3 demonstrates the vapor recompression process.

This layout of the vapor recompression process has some differences from the standard absorption process configuration, which are:

- Addition of a flash tank
- Addition of a compressor
- Reduction of reboiler load in the desorption column
- Needs more electricity to operate the compressor

The gray box in Fig. 3 is the recompression configuration, which incorporates a flash tank and a compressor. By flowing via a pressure relief valve, the pressure of the outflow flow from the bottom of the

desorption column is lowered. As a result of the pressure drop, some of the liquid condenses into steam. The vapor/liquid mixture enters the flash tank, separating it into vapor and liquid. Steam enters the compressor, condenses, and flows into the desorption column. The load on the reboiler in the desorption column is reduced, although the compressor imposes an additional demand on the unit. The remaining procedures are the same as in the standard configuration. In this configuration, although there is a reduction in energy consumption in the reboiler due to steam generation by the compressor, the capital cost increases as extra units are required compared to the standard configuration

2.4. Vapor recompression process with split-flow process configuration

This configuration is based on the combination of vapor recompression and split-flow processes to benefit from energy consumption reductions of both processes. In this configuration, the semi-lean amine is drawn from the desorption column before its flashing for vapor recompression. Fig. 4 demonstrates the vapor recompression process with a split-flow process configuration.

The difference between this configuration and the recompression configuration discussed in the previous section is the division of the lean steam into two parts before entering the flash tank. One of the streams enters the flash tank, similar to the recompression configuration. Still, the other stream, also called the semi-lean stream, enters directly into the primary heat exchanger of the unit. It is then sent to a cooler by a pump to reach the right temperature to enter the absorption column. The modification generates excess heat in the unit's primary heat exchanger, which affects the load on the desorption column's reboiler. This configuration needs less compressor power than the recompression configuration. Of course, equipment such as a pump, a cooler, and more piping in the absorption column is added to the configuration compared to the previous one, which should be considered from an economic point of view.

2.5. intercooling process configuration

The carbon dioxide absorption by MEA is an exothermic reaction that increases the solvent temperature passing through the absorber column. This phenomenon helps the kinetics of the reaction because of improved mass transfer coefficients and limits the capacity of solvent absorption due to the increased partial pressure of carbon dioxide. The intercooling process configuration can overcome absorption capacity

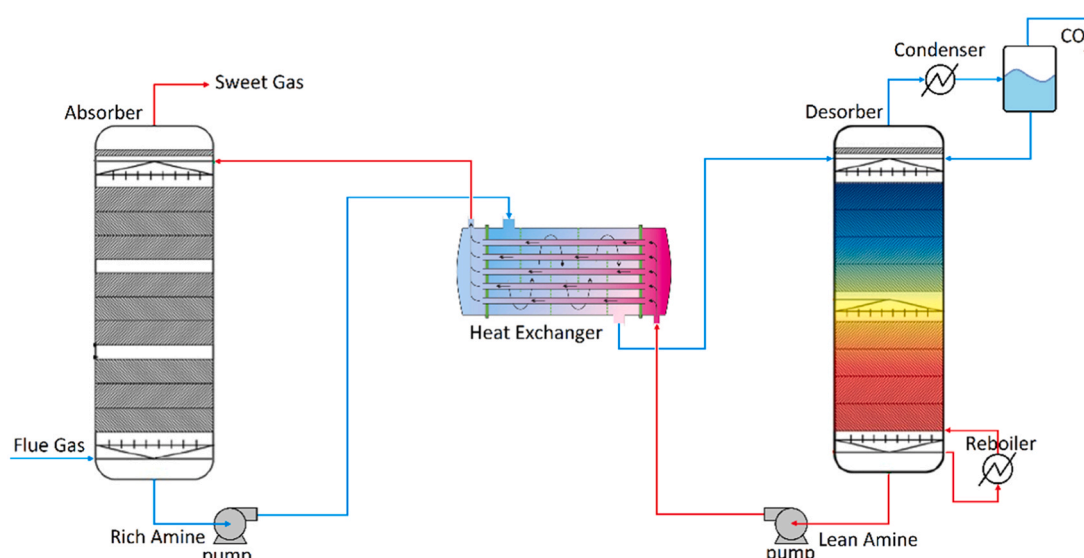


Fig. 2. Standard absorption process configuration.

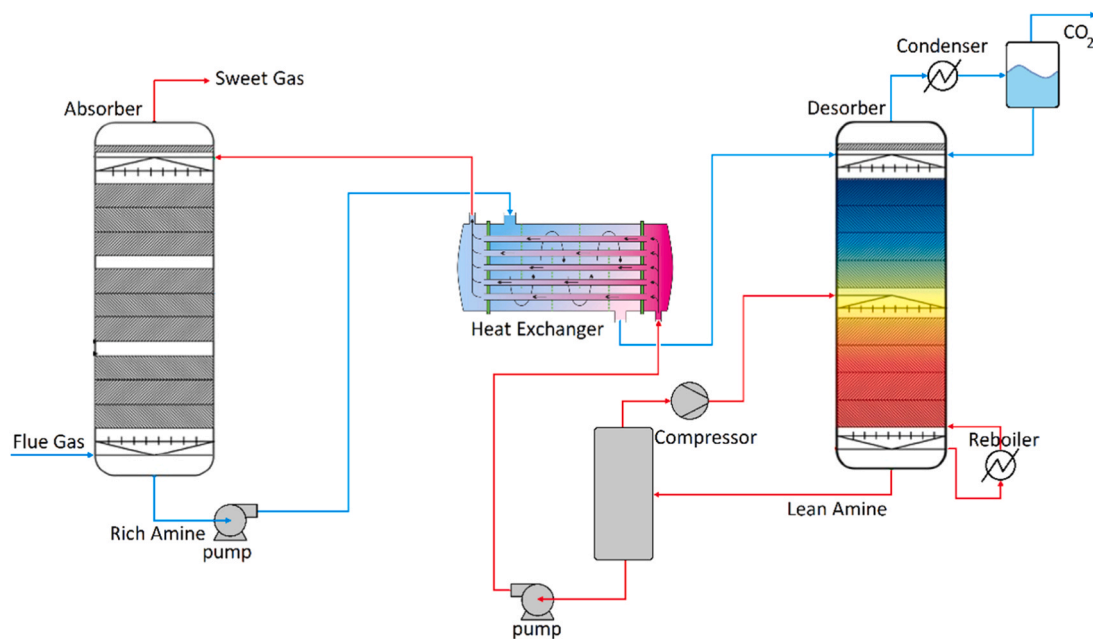


Fig. 3. Vapor recompression process configuration.

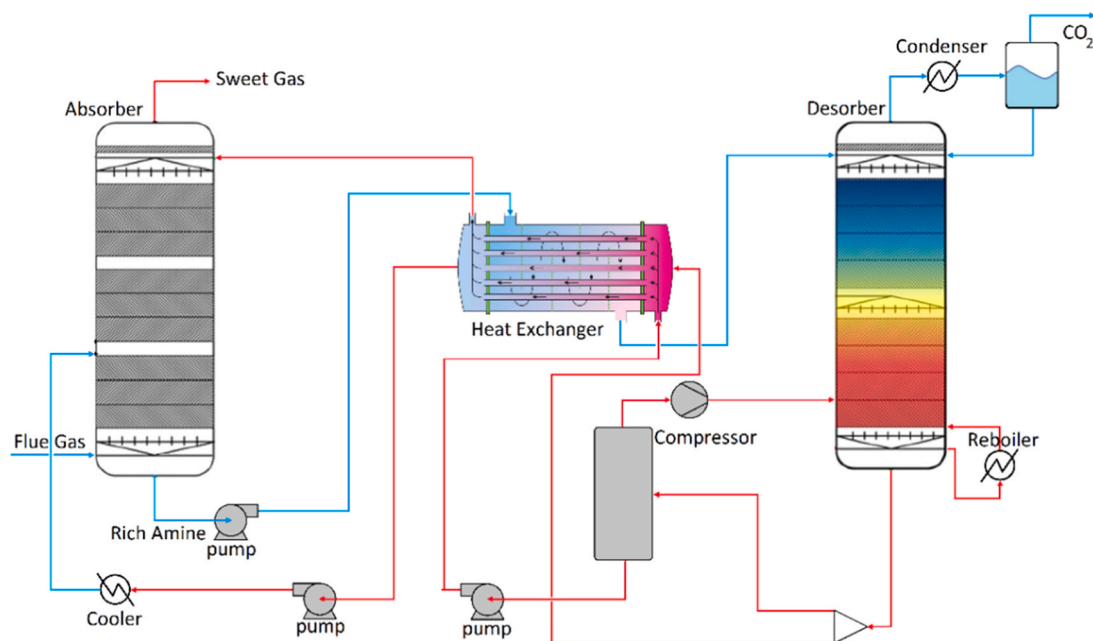


Fig. 4. The lean-split process with vapor recompression process configuration.

drawbacks due to increasing temperature through CO_2 loading increase in the rich solvent flow. The intercooling causes (Knudsen et al., 2011) a higher carbon dioxide cyclic carrying capacity, decreasing needed regeneration heat. Also, the intercooling process configuration can reduce the column height while keeping the removal efficiency of CO_2 at 85 % (Li et al., 2016a).

The reaction of carbon dioxide with an amine solvent is exothermic, thus increasing the temperature of the solvent absorbed throughout the absorption column. On the other hand, this increase in temperature is very suitable for the reaction because it improves the mass transfer coefficients. On the other hand, it limits the solvent absorption capacity due to the slight increase in carbon dioxide pressure at a given CO_2 loading. However, the intermediate cooling process can overcome these

problems by limiting the absorption capacity due to the rise in temperature along the column. This modification increases the carbon dioxide load in the rich solvent stream due to the low partial equilibrium pressure at low temperatures. Knudsen et al (Knudsen et al., 2011). examined the absorption column with intermediate cooling on a laboratory scale. They found that the carbon dioxide load increases in the rich stream, resulting in less required absorption energy. As shown in Fig. 5, a stream exits the column, is cooled by a coolant, and enters into the absorption column. The intercooling process is utilized to establish a uniform temperature in the absorber column, resulting in increased cooling duty and cooler component requirements for this task

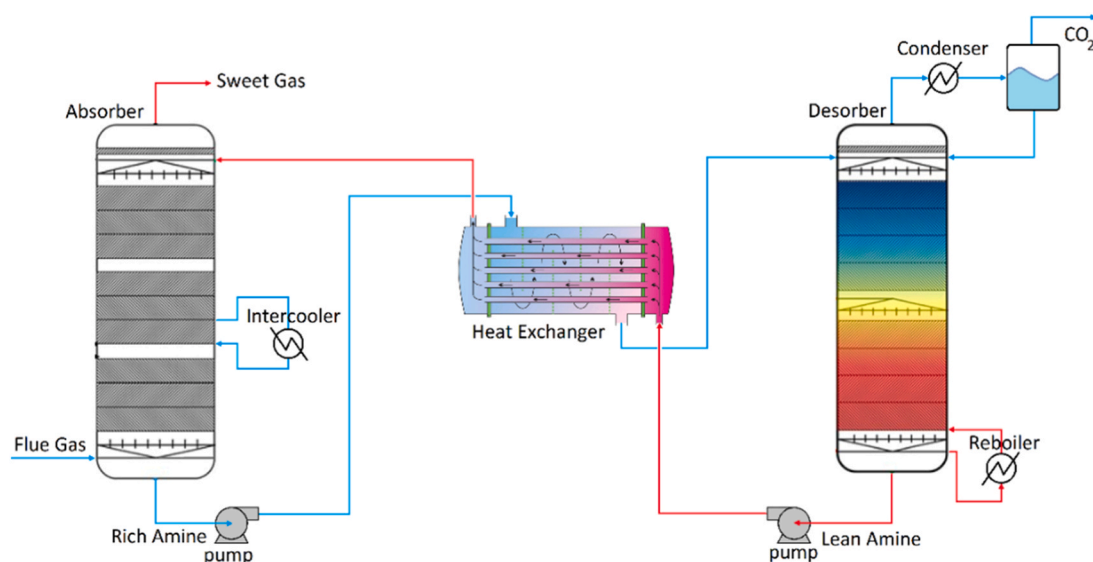


Fig. 5. Absorption column intercooling process configuration.

2.6. Rich-split configuration

The rich-split process configuration is a method to decrease the duty of the reboiler through the steam recovery generated in the desorber. The purpose of rich-flow splitting is to recover the energy of high-temperature vapor; simultaneously, the rich solvent is heated to release a portion of the carbon dioxide. The effectivity of this process was verified in process simulation (Lin et al., 2014; Li et al., 2015) and pilot plant trial (Cousins et al., 2012). The effectiveness of the rich-split configuration is considerably affected by the lean/rich heat exchanger efficiency. The elevated temperature along the desorber column results in increased water vapor pressure and decreased carbon dioxide vapor pressure. This phenomenon enhances the driving force of carbon dioxide desorption, which accelerates the rate of carbon dioxide desorption and lowers the heat consumption of solvent regeneration. The rich-split process configuration has three heat requirements: the heat of water vaporization, the heat of CO_2 desorption, and sensible heat (Li et al., 2016a). The heat of CO_2 desorption is unchanged due to the same rich solvent getting in the desorber and the corresponding CO_2 desorption rate. The rich-split configuration reduces the heat needed for water

vaporization, lowering the condenser and regeneration duty. On the other hand, the cold flow cools down the desorber, and a higher rate of sensible heat is needed to heat the split solvent; therefore, reboiler duty increases (Li et al., 2016a). So the optimum split fraction should be found to minimize the reboiler duty. Also, the heat exchanger temperature approach on the hot side affects the optimal split fraction.

The rich-split process configuration shown in Fig. 6 is an efficient modification to reduce the reboiler load while recovering the steam generated in the desorption column. The rich stream is divided into two parts, one stream similar to other configuration enters the primary heat exchanger of the unit, and the additional colder stream enters directly into the desorption column and recovers the energy remaining in the vapor produced in the desorption column, which moves to the top of the column, while the rich heated stream releases the carbon dioxide it contains (Van Wagener and Rochelle, 2011) (Li et al., 2015). The rich split process provides benefits by increasing solvent loading, reducing reboiler needs by decreasing the volume of streams and reducing condenser duty and stripper column requirements, making it advantageous compared to other configurations.

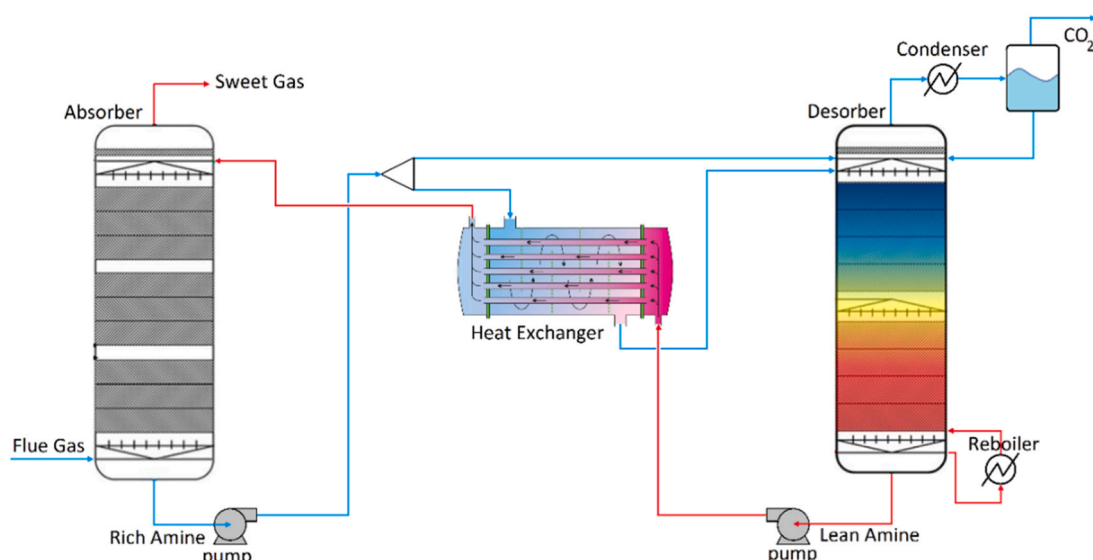


Fig. 6. Rich-split process configuration.

2.7. Combined configuration

The combined process was proposed to take advantage of the inter-heating, rich-split, and intercooling processes. In this configuration, the cooling duty is increased because of the intercooling process. The considerable size reduction of the absorber column can compensate for the elevated cooling duty. The operational conditions of this configuration are demonstrated in Fig. 7.

The combined configuration is designed to take advantage of the previous configurations, such as the absorption column with intermediate cooling, the rich flow division, and the desorption column with intermediate heating. According to Fig. 7, a stream exits from one of the stages of the desorption column and exchanges heat with the output stream from the bottom of the desorption column in a heat exchanger, and then the stream taken from the middle of the column returns to the column. Other operating conditions of the configuration are similar to the previous ones.

2.8. Rich-split with vapor recompression configuration

This configuration is a combination of rich-split and recompression processes. The goals of introducing this configuration are reboiler load reduction and heat utilization of vapor steam, condenser load reduction, and capital cost decrease of extra equipment in the combined configuration (Fig. 8).

Two types of energy utility are used in a carbon dioxide capture unit. Therefore, to compare different carbon capture configurations, there is a need to use the thermodynamic equivalent work definition. This definition uses a carbon capture unit that requires heat's potential for electricity generation to compute thermodynamic equivalent work. It is calculated using the below equation (Karimi et al., 2011):

$$W_{eq} = \left(Q_H \times \left(1 - \frac{T_c}{T_H} \right) \times \eta \right) + W_p + W_c \quad (1)$$

Q_H is the total consumed heat in the reboiler, η is the pump efficiency for electricity generation from the steam, which is assumed to be 75 %, and W_p and W_c are the needed power for the pump and compressor. The most critical parameters of carbon dioxide capture are carbon dioxide capture rate, MEA concentration, CO_2 loading of the lean solvent, the lean solvent temperature, and desorption column pressure. These parameters are studied as follows. The carbon dioxide capture rate is the most critical parameter for studying different configurations. Based on the operational policy of the capture unit, economic requirements, and

emission-related rules, the carbon dioxide capture rate is determined. Therefore, there is a need for techno-economic studies considering GHG emission rules in the future. In this study, carbon dioxide capture is assumed to be 85 %. The heat consumption in the reduction column reduces considerably by increasing the MEA concentration. The higher concentration of MEA leads to carbon dioxide capture performances such as reaction rate and carbon dioxide capture capacity per kg solvent. Also, higher carbon dioxide concentration leads to a lower solvent circulation rate in the carbon capture unit, directly affecting sensible heat reduction and, therefore, lower reboiler load.

By increasing MEA concentration from 25 wt% to 40 wt%, reboiler load decreases by 14 %. However, high MEA concentration due to the lack of oxygen leads to a higher MEA decomposition rate (Léonard et al., 2015). In this study, to achieve a proper balance between energy consumption and MEA decomposition rate, the concentration of MEA is assumed to be 30 %.

2.9. Lean solvent

According to the study by Abu-Zahra et al (Abu-Zahra et al., 2007), the optimum CO_2 loading of lean solvent is determined to be 0.3 for the MEA with 40 wt%. In another study by Alhajaj (Alhajaj et al., 2016), this parameter is reported to be 0.31. In another study, for MEA with 30 wt %, the CO_2 loading of the lean solvent was reported to be 0.25 for minimum reboiler heat load (Alie et al., 2005). Increasing the temperature of the lean solvent flow entering the absorption column would increase the reboiler heat load slightly. This increase is due to the lower capacity of the MEA solvent for absorption. On the other hand, the capture unit needs a lower cooling load for higher temperature conditions. As a result, the water consumption requirement in the unit for cooling would decline. This study assumes that the lean solvent entering the absorption column is 40 °C.

2.10. Stripper column pressure

The pressure increase of the desorption column has two advantages; It leads to heat demand reduction of water vaporization and increases in desorption column temperature. Therefore, the carbon dioxide separation becomes easier. In one study (Li et al., 2016a), operational pressure rise from 150 kPa to 275 kPa caused an 8 % decline in the heat consumption of the carbon dioxide capture process. On the other hand, a desorption column pressure increase has several disadvantages; temperature increases, the solvent decomposition rate increases, erosion

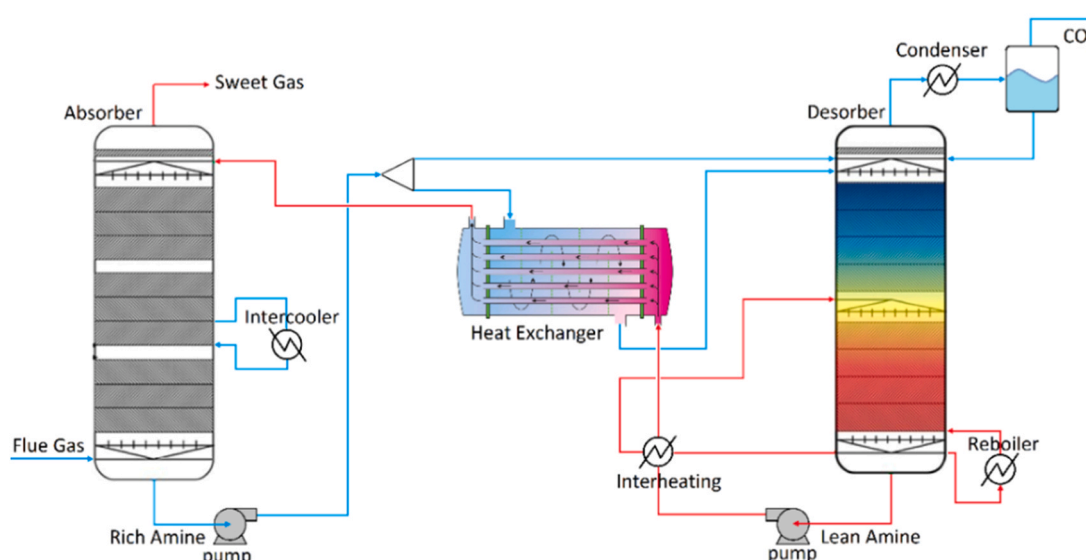


Fig. 7. Combined process configuration.

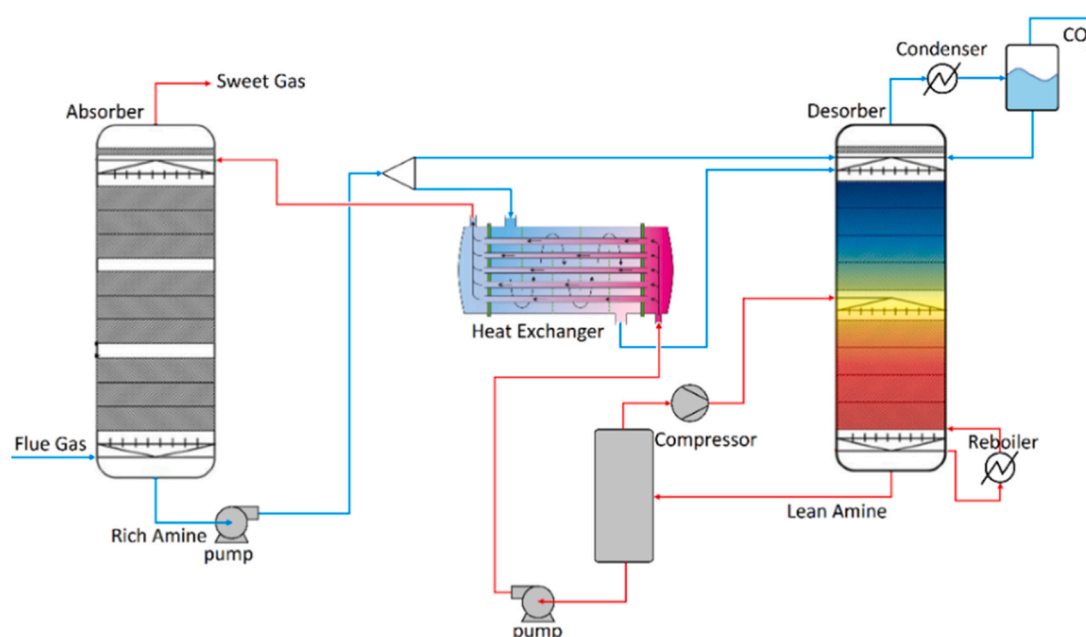


Fig. 8. Rich-split with vapor recompression process configuration.

problems, and higher maintenance and material costs. The pressure increase would also increase the capital cost of the desorption column. The optimum pressure for the desorption column is reported to be 2 bar in several studies (Li et al., 2016a; Amrollahi et al., 2012; Ziaii et al., 2011). The pressure of the absorption column is 1 bar plus the pressure drop in the column due to its packing.

2.11. Characteristics and simulation of different CO₂ capture process configurations

In this section, a different configuration presented in Section 2 is simulated. The characteristics and the results of each configuration simulation are presented and compared. The pressure drop in the heat exchangers and columns is neglected in the simulations.

2.11.1. Standard absorption process configuration

As can be seen in Fig. 8, the lean solvent flow enters the absorption column with 40 °C temperature and 100 kPa pressure. The number of the absorption column is 14. Then, the rich flow leaves the bottom of the absorption column at 42 °C temperature and 0.48 bar pressure. Then, it is pumped to the exchanger and enters the bottom of the desorption column with 104.4 °C temperature and 200 kPa pressure. The minimum temperature difference in the primary heat exchanger of the unit is assumed to be 10 °C. The lean solvent flow temperature after the heat exchanger is 52.17 °C. It should be cooled. Therefore, it enters a cooler, and its temperature will become 40 °C. Then, an amount of water and an MEA solvent is added to the flow. The solvent flow enters the column again.

The results of the simulation, such as reboiler load and the pumps' power consumption, are presented in Table 2.

2.11.2. Vapor recompression process configuration

After the departing reduction column, the lean solvent flow enters the throttling valve, and its pressure reaches 117 kPa. Then it enters the flash column. The outputs of the flash column enter the compressor, and its pressure reaches 200 kPa. The output gas of the compressor enters the 10th stage of the reduction column. The minimum temperature difference in the primary heat exchanger is 10 °C. The numbers of stages of the absorption column and reduction column are 16 and 10, respectively. Lean solvent flow and rich solvent flow loadings are 0.28 and 0.5,

respectively. As can be seen from Table 2, The Reboiler duty and the equivalent work are more than 17 % and 14 % lower than the reference configuration, respectively.

2.11.3. Vapor recompression process with split-flow process configuration

The main difference between this and the recompression configuration is splitting the lean solvent flow exiting the reduction column. 40 % of the flow enters the flash column and, like the recompression configuration, is compressed with a compressor and then enters the reduction column. 60 % of the lean solvent flow enters the heat exchanger, and its temperature is decreased to 40 °C and enters the absorption column. The minimum temperature difference in the primary heat exchanger is 5 °C. The primary purpose of this configuration was to decrease the compressor power consumption. Compressor power consumption of the lean-split process with vapor recompression process configuration is 45 % lower than the vapor recompression process configuration (Table 2). The reboiler duty of this configuration is 2 % and 20 % lower than the vapor recompression process configuration and the reference configuration, respectively.

2.11.4. Intercooling process configuration

In the intercooling process configuration, like other configurations, the lean solvent flow enters the absorption column with 40 °C temperature and 100 kPa pressure. In this configuration, the absorption column has 14 stages. In this study, the flow exited from the absorption column, cooled to 25 °C, and then entered the absorption column's last stage. The minimum temperature difference in the primary heat exchanger is 10 °C. The reboiler duty and equivalent work in Table 2 are 2.6 % and 2.8 % lower than the reference configuration. However, this configuration was not as successful as the vapor recompression process and the lean-split process with vapor recompression process configurations.

2.11.5. Rich-split process configuration

As was discussed earlier, to decrease the reboiler duty and heat recovery of generated vapor in the reduction column, the rich flow is split before entering the heat exchanger. 90 % of the flow enters the heat exchanger and the rest 10 % enters the reduction column directly. The exiting flow from the heat exchanger enters the second stage, and the split flow enters the first stage of the reduction column. With this modification, the duty of the reboiler decreases. The reboiler duty and

Table 2
Simulation results of seven different carbon capture process configuration.

Standard absorption	Vapor recompression		The lean-split process with vapor recompression		intercooling	rich-split	combined	rich-split with vapor recompression
Reboiler duty (MW)	251.8	214.3	210	245.3	234	201.8	193.8	
Reboiler energy consumption intensity (MJ.kg _{co₂} ⁻¹)	3.8	3.22	3.1	3.69	3.52	3.03	2.92	
Compressor power consumption (kW)	0	3670	2521	0	0	0	3900	
Pumps power consumption (kW)	491.1	294	453	495.3	491	490.3	486.8	
Equivalent work (MJ.kg _{co₂} ⁻¹)	2.13	1.867	1.81	2.07	1.98	1.70972	1.7	

equivalent work of the rich-split process configuration are 7 % lower than the same parameters in the reference configuration (Table 2).

2.11.6. Combined configuration

The combined configuration is a combination of intercooling, interheating, and rich-split processes. The lean solvent flow enters the absorption column like other configurations. The number of absorption columns is 18. The flow exits the sixteenth stage of the absorption column, is cooled to 25 °C, and enters the 18th stage of the absorption column. The rich solvent flow with 37 °C temperature split before entering the heat exchanger, and 90 % of it enters the heat exchanger, and the rest 10 % enters directly to the first stage of the reduction column. After the heat exchanger, the rich solvent flow enters the second stage of the heat exchanger. It leaves the third stage of the reduction column, transfers heat to lean solvent exiting the bottom of the reduction column, and enters the 10th stage of the reduction column at 110 °C temperature. The minimum temperature difference in the primary heat exchanger is 10 °C. As seen in Table 2, the reboiler’s duty has decreased to 201.8, which is 20 % lower than the base configuration, and the equivalent work is 24 % lower than the base configuration equivalent work.

2.11.7. Rich-split with vapor recompression configuration

The rich-split with vapor recompression configuration as the last configuration. The configuration is the combination of the rich-split solvent and vapor recompression configurations. 90 % of the rich solvent enters the heat exchanger, and the rest 10 % enters the first stage of the reduction column. The exiting lean solvent flow from the reduction column enters the throttling valve to 115 kPa and enters the flash column. The discharging liquid from the flash column enters the primary heat exchanger, and the discharging gas from the flash column enters the compressor. The discharging gas pressure is compressed to 200 kPa and enters the 10th stage of the reduction column. The minimum temperature difference in the primary heat exchanger is 5 °C. The simulated results in Table 2 show the rich-split with vapor recompression process configuration lower reboiler duty and equivalent work.

2.12. Economic model

One of the most critical goals in evaluating a carbon dioxide capture unit design is its analysis and upgrading from the economic viewpoint. In different configurations, pieces of equipment are added to reduce the reboiler’s energy consumption. The economic evaluation of adding these pieces of equipment to different configurations is performed in this study. Life cycle analysis was used for the economic analysis and evaluation of the configurations. In this method, the time value of money is considered. By taking into account the inflation rate, for costs and incomes, their values are assumed for the lifetime of the configurations (Kalogirou, 2014).

Life Cycle Cost (LCC) considers all costs and incomes of a system of its operating lifetime for calculating the system’s current value (net present value (NPV)). The time value of money is calculated based on the highest rate of interest rate in banks. In this method, the maintenance and operation costs of the system are calculated at the end of each year, and its NPV is calculated based on the below equation:

$$PW_n = \frac{A}{(1 + d)^n}$$
 (2)

A denotes total maintenance and operation costs at the end of each year, d is the inflation rate, and n is the system’s life in the year.

The main aim of this study is to compare different carbon dioxide capture with each other economically; hence due to the same carbon dioxide capture rate of all configurations, the income from CO₂ sale and exemption from GHG emissions taxes are not considered in this study. The below equation can be used to calculate the current capital costs of equipment:

$$C_{new} = C_{base} \left(\frac{Capacity_{new}}{Capacity_{Base}} \right)^{\alpha}$$

(3)

α is different for each piece of equipment, and its value can be found in the literature. The cost of equipment in the base year is found by multiplying the cost index by the base cost of the equipment.

$$C_2 = C_1 \left(\frac{I_2}{I_1} \right)$$

(4)

C denotes purchased cost, I is the cost index, and in this study, the Chemical Engineering Plant Cost Index (CEPCI) is the cost index (Lozowski, 2018). Indexes 1 and 2 denoted the equipment price in the reported year and required year, respectively. The economic functions of the equipment for carbon capture are presented in Table 3.

After determining the equipment purchase costs, the total capital investment cost is determined using Table 4. According to reference (Peters and Timmerhaus, 2002), direct and indirect costs are calculated according to the equipment purchase costs.

Table 5 calculates indirect costs such as engineering and construction costs.

The summation of direct and indirect costs results in fixed capital investment. Working investment is assumed to be 20 % of fixed capital investment—the summation of fixed capital investment and working investment results in total costs of investment (Peters and Timmerhaus, 2002).

$$FCI = DC + IDC$$

(5)

$$TCI = FCI + WI$$

(6)

Maintenance cost is assumed to be 2 % of the total investment cost. The consumed utility of the unit contains low-pressure steam, electricity, and cooling water. Steam costs are calculated by the below equation (Author1 et al., 53] </id><AuthGrp><Author><au> Department of Energy</au></AuthGrp>,</AuthGrp>).

$$C_f = a_f \times \frac{(H_{LP} - h_{bw})}{\eta_{boiler}}$$

(7)

C_f is the fuel cost per unit mass (USD/kg), a_f is the fuel cost per unit energy (USD/kJ), H_{LP} and h_{bw} are steam enthalpy and enthalpy of entering the water to the boiler (kJ/kg), and η_{boiler} is boiler efficiency.

$$C_{lp} = C_f \times (1 + 0.3)$$

(8)

Table 3
Cost functions of equipment (Li et al., 2016b; Mosaffa et al., 2016; Khoshgoftar Manesh et al., 2013).

Equipment name	Cost Function
Absorption column	$C_{absorber}(k\$) = 5235 \times \left(\frac{V_{new}}{V_{old}} \right)^{0.6}$
Absorption column packing	$C_{packing, absorber}(k\$) = 3823 \times \left(\frac{V_{new}}{V_{old}} \right)^{0.6}$
Desorption column	$C_{Desorber}(k\$) = 1423 \times \left(\frac{V_{new}}{V_{old}} \right)^{0.6}$
Desorption column packing	$C_{packing, Desorber}(k\$) = 915 \times \left(\frac{V_{new}}{V_{old}} \right)^{0.6}$
Heat Exchanger	$C_{Heat\ exchanger}(\$) = 1397 \times A_{heat\ transfer}^{0.89}$
Pump	$C_{pump}(\$) = 3540 \times W_{pump\ power}^{0.71}$
Compressor	$C_{Compressor}(k\$) = 538 \times 0.0063 \times \dot{m}_a \times \left(\frac{P_2}{P_1} \right)^{0.45} \times e^{0.45},$ $\dot{m}_a : \text{Gas mass flow rate}(kg \bullet S^{-1}), e = \frac{\eta_{is}}{1 - \eta_{is}}$
Flash column	$C_{Flash\ Tank}(\$) = 280.3 \times \dot{m}_{in, sep}^{0.67}, \dot{m} =$ $\text{input mass flow rate of flash tank}(kg.s^{-1})$
Throttling valve	$C_{valve}(\$) = 114.5 \times \dot{m}, \dot{m} :$ $\text{input mass flow rate of throttling valve}(kg.s^{-1})$

Table 4
Direct costs calculation of carbon capture unit (Peters and Timmerhaus, 2002).

Direct costs	The percentage range of capital costs	Assumed percentage of capital costs in this study
Equipment purchase	100	100
Equipment installation cost	25–55	50
Control and installation unit	8–50	20
Piping	20–80	40
Electrical systems	15–30	20
Service buildings	10–80	10
Landscaping	10–20	10
Service facilities	30–80	20
Land	4–8	5

Table 5
Indirect cost calculation of carbon capture unit (Peters and Timmerhaus, 2002).

Indirect costs	The percentage range of capital costs	Assumed percentage of capital costs in this study
Engineering	10	10
Construction	10	10
Miscellaneous and Contractor	17	17

The steam cost is calculated using the above equation. The consumed utility in the carbon dioxide capture unit is assumed based on the selling price of Mobin Petrochemical Co (Mobin Petrochemical Company, 2019). All the assumptions are presented in Table 6.

3. Result and discussion

3.1. Results of economic analysis of different carbon capture configurations

The purchased equipment costs are calculated based on Table 3 for each configuration. Then, using Table 4 and Table 5 data, each configuration’s direct and indirect costs are computed. Operational costs for each utility are calculated based on the consumed utility of each configuration using Table 6. Direct costs, indirect costs, fixed capital investment, and operational costs are calculated for each configuration as follows.

3.1.1. Standard absorption process configuration

The absorption column, desorption column, heat exchanger, reboiler, condenser, cooler, pump, and throttling valve are used in this configuration. This configuration’s direct and indirect costs are 79.912 and 29.967 million USD. This configuration’s total operational costs are calculated based on consumed utility and price. This configuration’s operating costs are reported in Table 7 for its base year.

Based on the total capital, maintenance, and operational costs, the total cost of the capture unit (NPV) is calculated to be 297.441 million

Table 6
Economic analysis assumptions for carbon dioxide capture (Li et al., 2016a; Firozeh Amini et al., 2015).

operating lifetime	25 years
Annual working hours	7200 hours
Repair costs	2 % of capital costs
Fuel inflation	5 %
Electricity inflation	6 %
Water inflation	10 %
MEA solvent price (USD/kg)	2.1
Steam price (USD/kg)	0.4
Water price (USD/m ³)	4
Electricity Price (¢/kWh)	7.5

Table 7

Operational costs of standard absorption process configuration.

	Standard absorption-1	Vapor recompression-2	The lean-split process with vapor recompression-3	Intercooling-4	Rich-split-5	Combined-6	Rich-split with vapor recompression-7
Utility type	Consumption amount						
Low-pressure steam (kg/s)	114.7	97.55	94.7	111.66	106.638	89.75	87.9
Water (kg/s)	3583.35	3583.75	1631.4255	3074.8	2710.55	2207	2137.16
MEA solvent (kg/s)	0.236	0.2131	0.1711	0.2115	0.2724	0.153	0.27
Electricity (kWh)	491.1	3964	2974	495.3	491	490.3	4386.8
Total operational costs	-	-	-	-	-	-	-
Utility type	Cost (Millions of USD annually)						
Low-pressure steam (kg/s)	12.95	11.01	10.69	12.61	12.04	10.13	9.93
Water (kg/s)	3.877	3.878	1.7654	3.327	2.933	2.39	2.312
MEA solvent (kg/s)	13.3812	12.08277	9.7	12	15.44	8.7	15.61
Electricity (kWh)	0.276	2.22975	1.672	0.278	0.276	0.275	2.46
Total operational costs	30.48	29.20	23.83	28.21	30.7	21.5	30.33

USD.

3.1.2. Vapor recompression process configuration

The absorption column, desorption column, heat exchanger, reboiler, condenser, cooler, pump, flash column, compressor, and throttling valve are used in this configuration. This configuration's direct and indirect costs are 77.850 and 29.193 million USD. This configuration's total operational costs are calculated based on consumed utility and price. This configuration's operational costs are reported in Table 7 for its base year. The total cost of this configuration for a 25-year operating lifetime is 295.7 million USD.

3.1.3. Vapor recompression process with split-flow process configuration

The absorption column, desorption column, heat exchanger, reboiler, condenser, cooler, pump, flash column, compressor, and throttling valve are used in this configuration. An extra cooler is the only difference between this configuration and the vapor recompression process. This configuration's direct and indirect costs are 92.107 and 34.540 million USD. This configuration's total operational costs are calculated based on consumed utility and price. This configuration's operational costs are reported in Table 7 for its base year. The total cost of this configuration for a 25-year operating lifetime is 281.652 million USD.

3.1.4. Absorption column with intercooling process configuration

The absorption column, desorption column, heat exchanger, reboiler, condenser, cooler, pump, and throttling valve are used in this configuration. The only difference between this configuration and the base configuration is an extra cooler. This configuration's direct and indirect costs are 82.818 and 31.057 million USD. This configuration's total operational costs are calculated based on consumed utility and price. This configuration's operational costs are reported in Table 7 for its base year. The total cost of this configuration for a 25-year operating lifetime is 291.61 million USD.

3.1.5. Rich-split process configuration

The absorption column, desorption column, heat exchanger, reboiler, condenser, cooler, pump, and throttling valve are used in this configuration. This configuration's direct and indirect costs are 80.824 and 30.309 million USD. This configuration's total operational costs are calculated based on consumed utility and price. This configuration's operational costs are reported in Table 7 for its base year. The total cost of this configuration for a 25-year operating lifetime is 290.7 million USD.

3.1.6. Combined configuration

The absorption column, desorption column, heat exchanger, reboiler, condenser, cooler, pump, and throttling valve are used in this configuration. In this configuration, an intercooler for the absorption column, and a heat exchanger as an interheater for the desorption column, are added to the base configuration. This configuration's direct and indirect costs are 102.095 and 38.285 million USD. This configuration's total operational costs are calculated based on consumed utility and price. This configuration's operational costs are reported in Table 7 for its base year. The total cost of this configuration for a 25-year operating lifetime is 287.3 million USD.

3.1.7. Rich-split with vapor recompression configuration

The absorption column, desorption column, heat exchanger, reboiler, condenser, cooler, pump, and throttling valve are used in this configuration. A compressor and a flash column are added to the base configuration in this configuration. This configuration's direct and indirect costs are 91.190 and 34.196 million USD. This configuration's total operational costs are calculated based on consumed utility and price. This configuration's operational costs are reported in Table 7 for its base year. The total cost of this configuration for a 25-year operating lifetime is 288.4 million USD.

3.2. Ranking of different configurations from a techno-economic point of view

In this research, Aspen HYSYS V10 was utilized to simulate multiple configurations in a post-combustion carbon capture unit aiming to reduce energy consumption compared to the standard configuration under steady-state conditions. The TOPSIS algorithm, within the framework of Multi-Criteria Decision Making (MCDM) methodology, was employed to rank and select the configurations based on their techno-economic aspects. Fig. 9 illustrates the comparison of various configurations in terms of steam, power, water, and MEA consumption per year. Configuration 6, which is a combined configuration identified as the best ranking configuration in the TOPSIS algorithm, is specified in this comparison in Fig. 9, showcasing the utility consumption reduction compared to the standard configuration and other configurations. The results depicted in this figure indicate that in the power consumption section, configuration 3 has the highest energy consumption due to the use of equipment such as a compressor for the heat pump function. Despite this increase compared to the standard configuration, configuration 3 effectively reduces steam usage and other aspects. Both configurations 3 and 6 demonstrate a reduction in steam usage compared to the standard configuration. In terms of water consumption for cooling duty, the study shows a minimal amount in configuration but utilizes

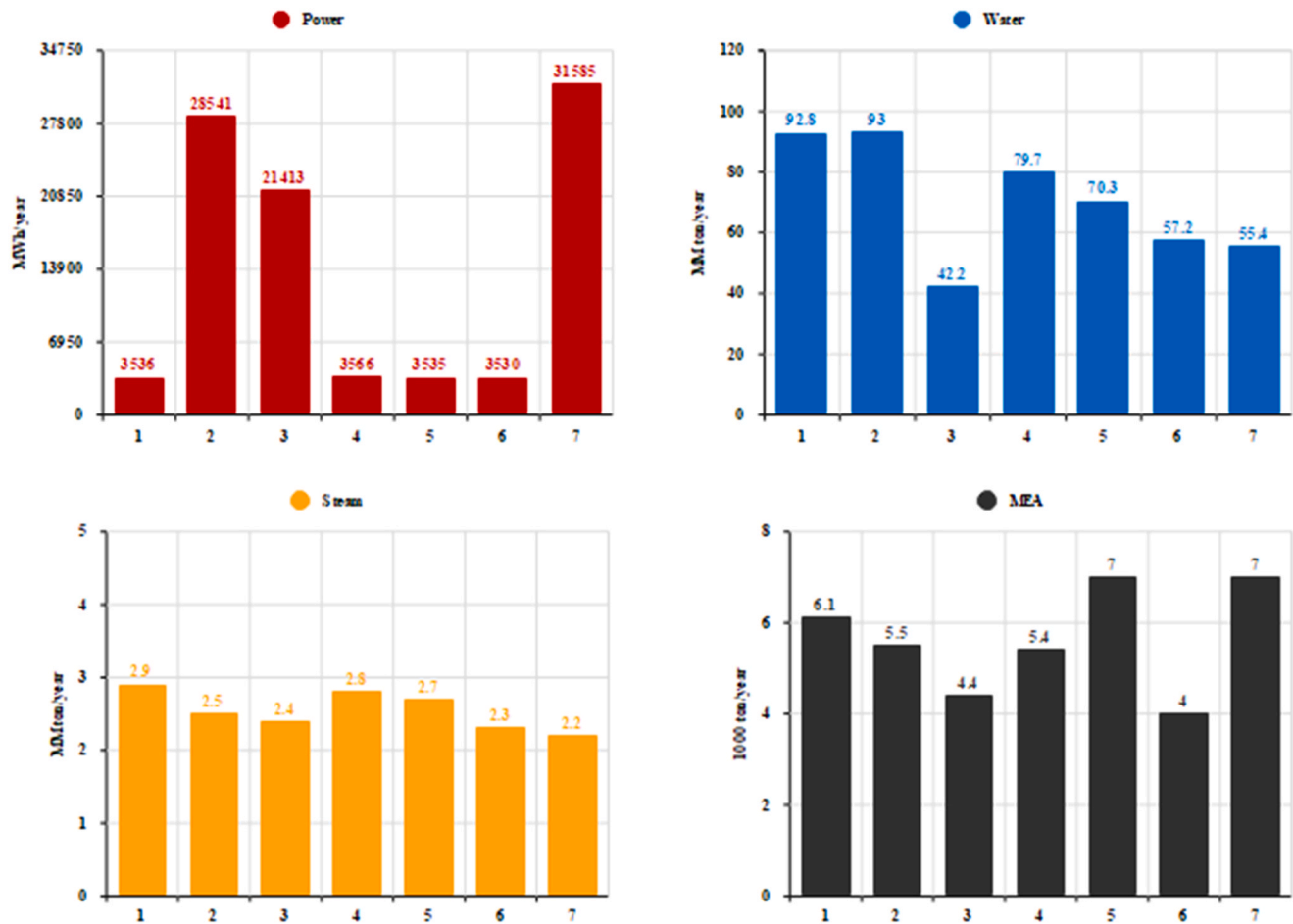


Fig. 9. Comparison various utility consumption in configurations (Standard absorption-1 Vapor recompression-2 The lean-split process with vapor recompression-3 Intercooling-4 Rich-split-5 Combined-6 Rich-split with vapor recompression-7).

split-flow cooling duty in the condenser stripper column, resulting in a 50 % reduction in this aspect. MEA consumption is heavily influenced by solvent regeneration thermal in the stripper column and absorption rate in the absorber column. In configuration 6, incorporating intercooling in the absorber column with split flow increases the temperature in the column and the absorption rate, leading to minimized amine usage in this configuration. While analyzing various configurations

based on utility usage, the ultimate decision rests on the economic total cost presented in Fig. 10.

Fig. 10 illustrates a comparison of various configurations based on the total cost of each configuration. The results indicate that in configuration 3, there is a reduction of about 16 million dollars in costs compared to the standard configuration. In both configuration 3 and 6, the total cost and utility consumption are lower than in other

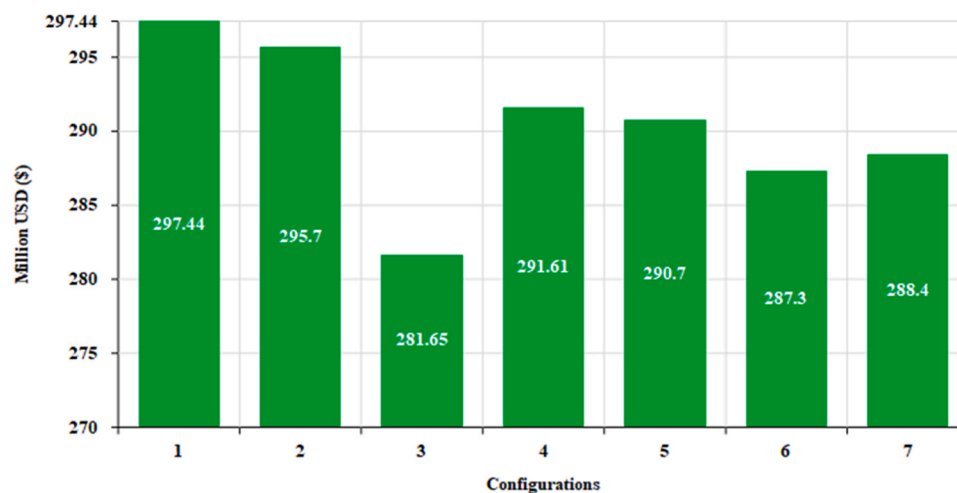


Fig. 10. Comparison various total cost in configurations (Standard absorption-1 Vapor recompression-2 The lean-split process with vapor recompression-3 Intercooling-4 Rich-split-5 Combined-6 Rich-split with vapor recompression-7).

configurations. Although configuration 3 involves using a compressor in conjunction with a heat pump, which affects steam usage in the reboiler stripper column, combining recompression configuration with lean split flow reduces costs by about 14 million dollars compared to using recompression configuration alone. This suggests that a combination of configurations could be beneficial for future studies. By combining configurations, the drawbacks of one configuration can be offset by the advantages of others. For example, in configuration 3, the lean split has an indirect effect on the power consumption of the configuration by reducing the volume of steam utilized.

One of the main goals of this research is to find the best configuration according to the defined criteria. The TOPSIS algorithm considers a wide range of economic parameters, such as utility costs, capital expenditures, operating costs, and maintenance expenses, providing a comprehensive assessment of costs compared to a standalone technoeconomic analysis. Utilizing the TOPSIS algorithm within the Multi-Criteria Decision Making (MCDM) methodology, one can effectively rank selection configurations for post-combustion carbon capture, determining the most suitable option based on predefined criteria. Decision-making can be done in one or more steps and provides the most appropriate option among those available according to the criteria. Each configuration is ranked using TOPSIS and based on the defined criteria (Asgharpour, 2016). TOPSIS method, as one of the best multi-criteria decision-making (MCDM) models, was proposed by Hwang and Yoon in 1981. This technique is based on the concept that the selected option should have the shortest distance from the ideal positive solution (best possible). In addition to the distance from the positive solution, in this method, the distance of the option from the negative ideal solution is also considered. This means that the selected option must have the shortest distance from the positive ideal solution and, simultaneously, the farthest distance from the negative ideal solution.

In this study, the criteria considered for the analysis by TOPSIS are the total cost of investment, consumption of low-pressure steam in the reboiler, electricity consumption in the Carbon Dioxide Absorption Unit, and water consumption of the unit. First, using the Shannon entropy method, Then, according to the TOPSIS algorithm, the values of C_i (*) for each layout were calculated, presented in Fig. 11.

Table 8 illustrates a comparison of various studies on absorption cases, showcasing the optimization configurations of PCC. The results indicate that the energy consumption in the reboiler of this study is lower compared to current studies. This reduced energy consumption is determined through an economic analysis for selection purposes.

The positive outcomes of this study and the feedback received

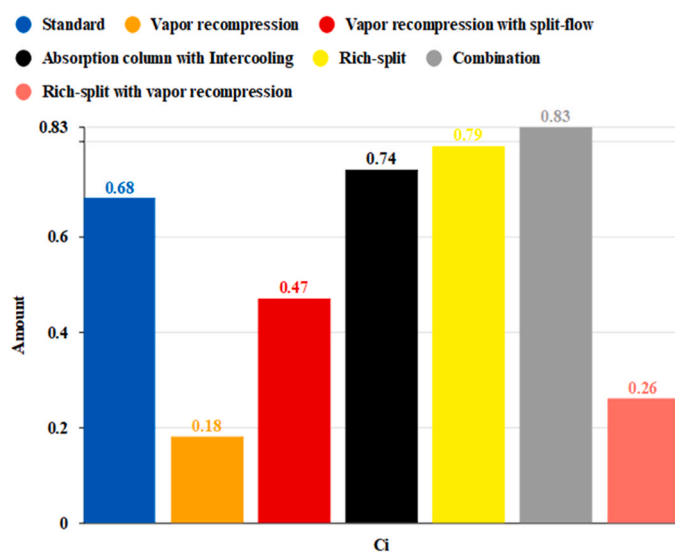


Fig. 11. Ranking of configurations.

Table 8

Comparison of various studies in absorption case.

Study	Reboiler energy (Gj/ton CO ₂)	Reference
This study	2.92	-
Ardali et al.	3.57	(Hosseini-Ardali et al., 2020)
Bravo et al.	4.56	(Bravo et al., 2021)
Dubois et al.	2.91	(Dubois and Thomas, 2018)

suggest that for future studies, it would be beneficial to simulate combinations of various approaches to enhance the advantages of the study. Additionally, incorporating renewable sources such as solar energy can improve the sustainability of the plant and enhance economic parameters by utilizing nature's fuel, the sun. To assess the negative environmental impacts of the plant resulting from the use of different configurations, environmental analyses can be employed. For instance, analyzing the impact of utilizing a recompression configuration involving component compressors and increased power consumption on the environmental effects can provide valuable insights. Advanced exergy analysis can also be utilized to identify the sources of irreversibility within components and facilitate measures to reduce this degradation, which indirectly impacts the plant's economic performance.

4. Conclusion

Seven alternative MEA absorption configurations were studied to propose the best configuration to lower the reboiler heat need. In the vapor recompression configuration, the pressure of the outflow flow at the bottom of the desorption column is reduced via a pressure relief valve. Some of the liquid condenses into steam due to the pressure drop. The mixture of vapor and liquid enters the flash tank, where it separates into vapor and liquid. Steam is introduced into the compressor, condensed, and then directed into the desorption column. As a result, the load on the reboiler in the desorption column is lowered while the compressor adds additional demand to the unit. In the vapor recompression process with the split flow, excess heat is generated in the unit's primary heat exchanger, affecting the load on the reboiler on the desorption column. Therefore, this configuration requires less compressor power than the recompression configuration. The absorption column with the intercooling configuration can compensate for the decrease in absorption capacity caused by increased temperature by increasing CO₂ loading in the rich solvent flow. Rich-flow splitting is used to recover energy from high-temperature vapor while simultaneously heating the rich solvent to release some of the carbon dioxide. The combined configuration was proposed to maximize the benefits of interheating, rich-split, and intercooling. The cooling duty is increased in this design due to the intercooling process. The significant size reduction of the absorber column compensates for the increased cooling duty. The Rich-split with vapor recompression configuration aims to reduce reboiler load and increase vapor steam heat usage, reducing condenser load and capital costs associated with additional equipment in the combined configuration. The results of the simulations show that rich-split with vapor recompression can lower reboiler heat duty better than other configurations. Still, considering the total cost of investment, power consumption, and water consumption of the configuration, it has become one of the worst. The results of the MCDM analysis show that combined and rich-split configurations have the best techno-economic performance among the presented configurations.

CRedit authorship contribution statement

Mohsen Salimi: Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Majid Amidpour:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Vahid Karimi:** Writing – review & editing,

Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Farzin Hosseinfard:** Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Abu-Zahra, M.R.M., Schneiders, L.H.J., Niederer, J.P.M., Feron, P.H.M., Versteeg, G.F., 2007. CO₂ capture from power plants. Part I. A parametric study of the technical performance based on monoethanolamine. *Int. J. Greenh. Gas. Control* vol. 1 (1), 37–46. [https://doi.org/10.1016/S1750-5836\(06\)00007-7](https://doi.org/10.1016/S1750-5836(06)00007-7).
- Ahn, H., Luberti, M., Liu, Z., Brandani, S., 2013. Process simulation of aqueous MEA plants for postcombustion capture from coal-fired power plants. *Energy Procedia* vol. 37, 1523–1531. <https://doi.org/10.1016/j.egypro.2013.06.028>.
- Akbari, M., Memarian, H., Neamatollahi, E., Jafari Shalamzari, M., Alizadeh Noughani, M., Zakeri, D., 2021. Prioritizing policies and strategies for desertification risk management using MCDM–DPSIR approach in northeastern Iran. *Environ. Dev. Sustain.* vol. 23 (2) <https://doi.org/10.1007/s10668-020-00684-3>.
- Alhajaj, A., Mac Dowell, N., Shah, N., 2016. A techno-economic analysis of post-combustion CO₂ capture and compression applied to a combined cycle gas turbine: part I. A parametric study of the key technical performance indicators. *Int. J. Greenh. Gas. Control* vol. 44, 26–41. <https://doi.org/10.1016/j.jggc.2015.10.022>.
- Alie, C., Backham, L., Croiset, E., Douglas, P.L., 2005. Simulation of CO₂ capture using MEA scrubbing: a flowsheet decomposition method. *Energy Convers. Manag.* vol. 46 (3), 475–487. <https://doi.org/10.1016/j.enconman.2004.03.003>.
- Alsalem, M.A., et al., 2022b. Multi-criteria decision-making for coronavirus disease 2019 applications: a theoretical analysis review. *Artif. Intell. Rev.* vol. 55 (6) <https://doi.org/10.1007/s10462-021-10124-x>.
- Alsalem, M.A., et al., 2022a. Rise of multiattribute decision-making in combating COVID-19: a systematic review of the state-of-the-art literature. *Int. J. Intell. Syst.* vol. 37 (6) <https://doi.org/10.1002/int.22699>.
- Amrollahi, Z., Ystad, P.A.M., Ertesvåg, I.S., Bolland, O., 2012. Optimized process configurations of post-combustion CO₂ capture for natural-gas-fired power plant - Power plant efficiency analysis. *Int. J. Greenh. Gas. Control* vol. 8. <https://doi.org/10.1016/j.jggc.2012.01.005>.
- Asgharpour, M.J., 2016. *Multiple Criteria Decision Making*, 1st ed. University of Tehran press, Tehran.
- Bravo, J., Drapanauskaite, D., Sarunac, N., Romero, C., Jesikiewicz, T., Baltrusaitis, J., 2021. Optimization of energy requirements for CO₂ post-combustion capture process through advanced thermal integration. *Fuel* vol. 283. <https://doi.org/10.1016/j.fuel.2020.118940>.
- Choi, J., et al., 2019. Process design and optimization of MEA-based CO₂ capture processes for non-power industries. *Energy* vol. 185. <https://doi.org/10.1016/j.energy.2019.07.092>.
- Cousins, A., Cottrell, A., Lawson, A., Huang, S., Feron, P.H.M., 2012. Model verification and evaluation of the rich-split process modification at an Australian-based post combustion CO₂ capture pilot plant. *Greenh. Gases Sci. Technol.* vol. 2 (5), 329–345. <https://doi.org/10.1002/ghg.1295>.
- Cousins, A., Wardhaugh, L.T., Feron, P.H.M., 2011a. A survey of process flow sheet modifications for energy efficient CO₂ capture from flue gases using chemical absorption. *Int. J. Greenh. Gas. Control* vol. 5 (4), 605–619. <https://doi.org/10.1016/j.jggc.2011.01.002>.
- Cousins, A., Wardhaugh, L.T., Feron, P.H.M., 2011b. Preliminary analysis of process flow sheet modifications for energy efficient CO₂ capture from flue gases using chemical absorption. *Chem. Eng. Res. Des.* vol. 89 (8), 1237–1251. <https://doi.org/10.1016/j.cherd.2011.02.008>.
- Department of Energy, “How to Calculate the True Cost of Steam.”
- Depledge, J., 2022. The ‘top-down’ Kyoto Protocol? Exploring caricature and misrepresentation in literature on global climate change governance. *Int. Environ. Agreem. Polit. Law Econ.* vol. 22 (4) <https://doi.org/10.1007/s10784-022-09580-9>.
- Dubois, L., Thomas, D., 2018. Comparison of various configurations of the absorption-regeneration process using different solvents for the post-combustion CO₂ capture applied to cement plant flue gases. *Int. J. Greenh. Gas. Control* vol. 69 (December 2017), 20–35. <https://doi.org/10.1016/j.jggc.2017.12.004>.
- M.K. Firozeh Amini, Mohammadali Shafizadeh, Lida Saber Fattahi, Mostafa Tavanpour, Pantea Soleimanpour, Majid Farmad, Nasrin Golghahremani, Iran’s energy balance sheet 2015. Tehran, 2015.
- Grennfelt, P., Englerlyd, A., Forsius, M., Hov, Ø., Rodhe, H., Cowling, E., 2020. Acid rain and air pollution: 50 years of progress in environmental science and policy. *Ambio* vol. 49 (4). <https://doi.org/10.1007/s13280-019-01244-4>.
- Hosseini-Ardali, S.M., Hazrati-Kalbibaki, M., Fattahi, M., Lezsovis, F., 2020. Multi-objective optimization of post combustion CO₂ capture using methyldiethanolamine (MDEA) and piperazine (PZ) bi-solvent. *Energy* vol. 211. <https://doi.org/10.1016/j.energy.2020.119035>.
- Hosseinfard, F., Hosseinpour, M., Salimi, M., Amidpour, M., 2024a. Optimizing post-combustion carbon capture: A comprehensive 3E analyses of energy, exergy, and exergoeconomic aspects with solar integration. *Sustain. Energy Technol. Assess.* vol. 62 (October 2023), 103626 <https://doi.org/10.1016/j.seta.2024.103626>.
- Hosseinfard, F., Hosseinpour, M., Salimi, M., Amidpour, M., 2024b. Toward a greener future: Solar solutions for industrial carbon capture (Mar.). *Sol. Energy* vol. 271, 112436. <https://doi.org/10.1016/j.solener.2024.112436>.
- Ismael, S.F., et al., 2023. Toward sustainable transportation: a pavement strategy selection based on the extension of dual-hesitant fuzzy multicriteria decision-making methods. *IEEE Trans. Fuzzy Syst.* vol. 31 (2) <https://doi.org/10.1109/TFUZZ.2022.3168050>.
- Jung, J., Jeong, Y.S., Lim, Y., Lee, C.S., Han, C., 2013. Advanced CO₂ capture process using MEA scrubbing: Configuration of a split flow and phase separation heat exchanger. *Energy Procedia* vol. 37, 1778–1784. <https://doi.org/10.1016/j.egypro.2013.06.054>.
- Kalogirou, S.A., 2014. *Solar Energy Engineering: Processes and Systems: Second Edition*. Elsevier/Academic Press doi: 10.1016/C2011-0-07038-2.
- Karimi, M., Hillestad, M., Svendsen, H.F., 2011. Capital costs and energy considerations of different alternative stripper configurations for post combustion CO₂ capture. *Chem. Eng. Res. Des.* vol. 89 (8) <https://doi.org/10.1016/j.cherd.2011.03.005>.
- Karimi, M., Hillestad, M., Svendsen, H.F., 2012. Positive and negative effects on energy consumption by interheating of stripper in CO₂ capture plant. *Energy Procedia* vol. 23, 15–22. <https://doi.org/10.1016/j.egypro.2012.06.066>.
- KEY WORLD ENERGY STATISTICS. IEA, 2019.
- Khoshtoghtar Manesh, M.H., Abadi, S.K., Amidpour, M., Hamed, M.H., 2013. A new targeting method for estimation of cogeneration potential and total annualized cost in process industries. *Chem. Eng. Res. Des.* vol. 91 (6), 1039–1049. <https://doi.org/10.1016/j.cherd.2012.12.002>.
- Knudsen, J.N., Andersen, J., Jensen, J.N., Biede, O., 2011. Evaluation of process upgrades and novel solvents for the post combustion CO₂ capture process in pilot-scale. *Energy Procedia* vol. 4, 1558–1565. <https://doi.org/10.1016/j.egypro.2011.02.025>.
- Kohl Richard Nielsen, Arthur L., 1997. *Gas Purification*. Gulf Professional Publishing.
- Lars Erik, Ø., 2012. “Remov. CO₂ Exhaust Gas.” 165.
- Le Mouellec, Y., Neveux, T., Al Azki, A., Chikukwa, A., Hoff, K.A., 2014. Process modifications for solvent-based post-combustion CO₂ capture. *Int. J. Greenh. Gas. Control* vol. 31, 96–112. <https://doi.org/10.1016/j.jggc.2014.09.024>.
- Léonard, G., Crosset, C., Toye, D., Heyen, G., 2015. Influence of process operating conditions on solvent thermal and oxidative degradation in post-combustion CO₂ capture. *Comput. Chem. Eng.* vol. 83, 121–130. <https://doi.org/10.1016/j.compchemeng.2015.05.003>.
- Li, K., et al., 2016a. Systematic study of aqueous monoethanolamine-based CO₂ capture process: model development and process improvement. *Energy Sci. Eng.* vol. 4 (1) <https://doi.org/10.1002/ese3.101>.
- Li, K., Leigh, W., Feron, P., Yu, H., Tade, M., 2016b. Systematic study of aqueous monoethanolamine (MEA)-based CO₂ capture process: Techno-economic assessment of the MEA process and its improvements. *Appl. Energy* vol. 165, 648–659. <https://doi.org/10.1016/j.apenergy.2015.12.109>.
- Li, K., Yu, H., Feron, P., Tade, M., Wardhaugh, L., 2015. Technical and energy performance of an advanced, aqueous ammonia-based CO₂ capture technology for a 500 MW coal-fired power station. *Environ. Sci. Technol.* vol. 49 (16), 10243–10252. <https://doi.org/10.1021/acs.est.5b02258>.
- Lin, Y.J., Madan, T., Rochelle, G.T., 2014. Regeneration with rich bypass of aqueous piperazine and monoethanolamine for CO₂ capture. *Ind. Eng. Chem. Res.* vol. 53 (10), 4067–4074. <https://doi.org/10.1021/ie403750s>.
- Liu, J., 2020. Investigation into an energy-saving mechanism for advanced stripper configurations in post-combustion carbon capture. *J. Taiwan Inst. Chem. Eng.* vol. 000 (2019) <https://doi.org/10.1016/j.jtice.2019.12.012>.
- Liu, W., et al., 2024. Thermodynamic analysis of adsorption carbon capture from limiting cycle to heat pump assisted cycle (Mar.). *Energy* vol. 291, 130299. <https://doi.org/10.1016/j.energy.2024.130299>.
- D. Lozowski, “The Chemical Engineering Plant Cost Index.” 2018.
- Massam, B.H., 1988. Multi-criteria decision making (MCDM) techniques in planning. *Prog. Plan.* vol. 30 (PART 1) [https://doi.org/10.1016/0305-9006\(88\)90012-8](https://doi.org/10.1016/0305-9006(88)90012-8).
- Milhorance, C., Sabourin, E., Chechi, L., Mendes, P., 2022. The politics of climate change adaptation in Brazil: framings and policy outcomes for the rural sector. *Env. Polit.* vol. 31 (2) <https://doi.org/10.1080/09644016.2021.1907097>.
- Mobin Petrochemical Company, Financial Statements of Mobin Petrochemical Company, 2019.
- Mohammed, A., Harris, I., Govindan, K., 2019. A hybrid MCDM-FMOO approach for sustainable supplier selection and order allocation. *Int. J. Prod. Econ.* vol. 217 <https://doi.org/10.1016/j.ijpe.2019.02.003>.
- Mosaffa, A.H., Farshi, L.G., Infante Ferreira, C.A., Rosen, M.A., 2016. Exergoeconomic and environmental analyses of CO₂/NH₃ cascade refrigeration systems equipped with different types of flash tank intercoolers. *Energy Convers. Manag.* vol. 117, 442–453. <https://doi.org/10.1016/j.enconman.2016.03.053>.
- Oexmann, J., Kather, A., 2010. Minimising the regeneration heat duty of post-combustion CO₂ capture by wet chemical absorption: the misguided focus on low heat of absorption solvents. *Int. J. Greenh. Gas. Control* vol. 4 (1), 36–43. <https://doi.org/10.1016/j.jggc.2009.09.010>.
- Oh, S.Y., Binns, M., Cho, H., Kim, J.K., 2016. Energy minimization of MEA-based CO₂ capture process. *Appl. Energy* vol. 169, 353–362. <https://doi.org/10.1016/j.apenergy.2016.02.046>.

- Peters, M., Timmerhaus, K., 2002. *Plant design and economics for chemical engineers*, 5th ed. McGraw Hill.
- Qu, Y., Liu, Y., Jevrejeva, S., Jackson, L.P., 2020. Future sea level rise along the coast of China and adjacent region under 1.5 °C and 2.0 °C global warming (Sep.). *Adv. Clim. Chang. Res.* vol. 11 (3), 227–238. <https://doi.org/10.1016/J.ACCRE.2020.09.001>.
- Soepyan, F.B., et al., 2024. Optimization of a natural gas power plant with membrane and solid sorbent carbon capture systems (Mar.). *Carbon Capture Sci. Technol.* vol. 10, 100165. <https://doi.org/10.1016/J.CCST.2023.100165>.
- Sotiropoulou, K.F., Vavatsikos, A.P., 2021. Onshore wind farms GIS-assisted suitability analysis using PROMETHEE II. *Energy Policy* vol. 158. <https://doi.org/10.1016/j.enpol.2021.112531>.
- Sun, R.S., Gao, X., Deng, L.C., Wang, C., 2022. Is the Paris rulebook sufficient for effective implementation of Paris agreement? *Adv. Clim. Chang. Res.* vol. 13 (4) <https://doi.org/10.1016/j.accre.2022.05.003>.
- Vafaei, N., Ribeiro, R.A., Camarinha-Matos, L.M., 2021. Assessing normalization techniques for simple additive weighting method. *Procedia Comput. Sci.* <https://doi.org/10.1016/j.procs.2022.01.156>.
- Van Wagener, D.H., Rochelle, G.T., 2011. Stripper configurations for CO₂ capture by aqueous monoethanolamine and piperazine. *Energy Procedia* vol. 4, 1323–1330. <https://doi.org/10.1016/j.egypro.2011.01.190>.
- Zhou, J., et al., 2023. Performance assessment and system optimization on supercritical CO₂ double-path recompression coal-fired combined heat and power plants with MEA-based post-combustion CO₂ capture (Mar.). *Energy* vol. 267, 126539. <https://doi.org/10.1016/J.ENERGY.2022.126539>.
- Ziaii, S., Rochelle, G.T., Edgar, T.F., 2011. Optimum design and control of amine scrubbing in response to electricity and CO₂ prices. *Energy Procedia* vol. 4, 1683–1690. <https://doi.org/10.1016/j.egypro.2011.02.041>.
- Ziemele, J., Gendelis, S., Dace, E., 2023. Impact of global warming and building renovation on the heat demand and district heating capacity: Case of the city of Riga. *Energy* vol. 276. <https://doi.org/10.1016/j.energy.2023.127567>.